Chapter 10. Functions in Python

A **function** is a reusable block of code that performs a specific task. Instead of writing the same code again and again, we write it once inside a function and call it whenever needed.

Basic idea

Function = reusable block of code

Simple flow

Define function
  |
  v
Call function
  |
  v
Function code runs
  |
  v
Result/output is produced

10.1 What is a Function?

Syntax

```python
def function_name():
    statement
```

1. A function groups related code together.
2. A function runs only when it is called.
3. Functions help avoid repeated code.
4. Functions make code clean, reusable, and easy to understand.
5. Python has built-in functions and user-defined functions.

Built-in vs User-defined Functions

Type	Meaning	Example
Built-in function	Already provided by Python	print(), len(), type()
User-defined function	Created by programmer	def greet():

CodeWithNishchal

Flow Chart

Start
 |
 v
Function is defined
 |
 v
Function is called
 |
 v
Function body executes
 |
 v
Program continues

PYTHON CODE

```
def greet():
    print("Hello, welcome to Python")


greet()
```

Output: Hello, welcome to Python

10.2 Defining Functions

Syntax

```
Python

def function_name():
    statement
```

1. def is used to define a function.
2. The function name comes after def.
3. Parentheses () are required.
4. A colon : marks the start of the function body.
5. The function body must be indented.

```
def       function_name      ()       :
 |            |             |        |
keyword     name         parentheses  colon
```

CodeWithNishchal

```
PYTHON CODE
def say_hello():
    print("Hello")
```

This only defines the function. It does not run yet. To run it, we must call it.

say_hello()

Output: Hello

Empty Function

If we want to create a function but write logic later, use pass.

```
PYTHON CODE
def future_function():
    pass
```

10.3 Calling Functions

Syntax

function_name()

1. Calling a function means executing it.
2. A function can be called once or many times.
3. The function body runs every time the function is called.
4. If a function is defined but never called, its code will not execute.

Example 1

```
PYTHON CODE
def greet():
    print("Hello")


greet()
greet()
greet()
```

Output:Hello
Hello
Hello

Code With Nishchal

10.4 Parameters

A **parameter** is a variable written inside the function definition. It receives values when the function is called.

Syntax

```
PYTHON CODE
def function_name(parameter):
    statement
```

Explanation

1. Parameters make functions flexible.
2. Parameters allow us to send data into a function.
3. A function can have one or more parameters.
4. Parameters are written inside parentheses during function definition.

Flow Chart

Function definition has parameter
  |
  v
Function call sends value
  |
  v
Parameter receives value
  |
  v
Function uses that value

Example 1

```
PYTHON CODE
def greet(name):
    print("Hello", name)
greet("Aman")
```

Output:
Hello Aman
Here, name is a parameter.

CodeWithNishchal